

SE 2226 Test Plan

Project: D&R E-Commerce Web Application & Library-DB-Express Testing

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1 Context of testing

1.1 Project/test levels/test types

This test plan is prepared for the term project to be carried out within the scope of the Software Quality Assurance and Testing course. Various levels and types of tests will be applied on two different software architectures.

- **D&R Web Application:** Will be evaluated at the System Testing level. Functional Testing, GUI (Graphical User Interface) Testing, and Usability Testing will be applied as test types.
- **Library-DB-Express Application:** Will be evaluated at the code level through Unit Testing and Mutation Testing.

1.2 Test items

There are two main items to be tested:

- **D&R Web Platform:** Turkey's leading book and hobby e-commerce platform (<https://www.dr.com.tr>). Tests will be conducted on the production version of the application, restricted strictly to guest user operations.
- **Library-DB-Express:** An open-source Node.js/Express based local project for book database operations, provided within the course scope (Version: Master Branch).

1.3 Test scope

This section details the features to be tested and the features excluded from the testing scope.

In-Scope Features:

- **For D&R:** Homepage search bar functionality, category and price filtering algorithms on the search results page, consistency of information on the product detail page, and add-to-cart/update-cart functions.
- **For Library-DB-Express:** Accuracy of data models within the source code and unit testing coverage of API endpoints.

Out-of-Scope Features:

- User login, registration, and payment (credit card/3D Secure) steps on the D&R website are excluded from the scope due to security firewalls (CAPTCHA) and bot protection mechanisms.
- Database and back-end testing of the D&R platform are excluded (due to lack of access).

1.4 Test basis

The design and execution of the tests will be based on the following resources:

- SE 2226 Course Syllabus and laboratory guidelines.
- ISO/IEC/IEEE 29119-2 Software Testing Standards.
- The expected User Experience (UX) workflows of the D&R web platform.

2 Assumptions and constraints

Assumptions and constraints that may affect the progress of the project during the testing process are listed below.

Assumptions:

- The user interface (DOM structure and XPath addresses) of the D&R web platform will not change with a major update while the test automation is being coded.
- It is assumed that all necessary tools (IntelliJ, Java, Node.js) will work flawlessly on laboratory computers and team members' personal computers.
- It is assumed that D&R servers will not perceive the automated test requests as bot attacks, thanks to the deliberate delay intervals (`Thread.sleep`) we will implement during Selenium automation.

Constraints:

- **Time Constraint:** All test planning, automation coding, and reporting must be completed before the laboratory presentation date on May 4th.
- **Budget and Tool Constraint:** Only open-source and free versions of tools (Selenium IDE, JUnit6, PIT Plugin) will be used throughout the testing process.
- **Access Constraint:** Only Black-box testing can be performed on the D&R platform since there is no access to its source code.

3 Test strategy

3.1 General

A hybrid testing approach will be adopted in this project. External Black-box tests simulating user behaviors will be executed on D&R using Selenium, while internal White-box tests validating the internal logic of the system will be written using JUnit on Library-DB-Express.

3.2 Test levels

- **D&R:** System Testing (Modular system validation without end-to-end automation).
- **Library-DB-Express:** Unit Testing and Integration Testing.

3.3 Test types

The test types to be executed are:

- Functional Testing

- GUI Testing
- Mutation Testing
- Performance Testing (Light Load Testing with JMeter - For extra credit)

3.4 Test deliverables

The project deliverables will include the following documents and outputs:

- Test Plan (This document).
- Test Case Specification (Equivalence, Boundary tables, etc.).
- Automation Source Codes (.java files and Selenium IDE .side files).
- Test Completion Report.

3.5 Test design techniques

This section details the test design techniques to be used and their application examples on D&R.

1. Equivalence Class Partitioning:

Will be applied to the "Minimum - Maximum Price" search function in the D&R book filtering section.

- **Valid Class 1:** 10 TL - 50 TL (Expected: Books within the relevant range are listed).
- **Invalid Class 1:** -10 TL - 50 TL (Expected: Warning that negative values cannot be entered).
- **Invalid Class 2:** 100 TL - 50 TL (Expected: Error stating Min value cannot be greater than Max value).

2. Boundary Value Analysis:

Will be tested on the maximum product (stock) limit that can be added to the cart. Assuming a maximum of 10 copies of a book can be added:

- **Lower Boundary:** Attempting to add 0 items (Cart is not updated).
- **Valid Values:** Adding 1, 9, 10 items (Cart is updated successfully).
- **Upper Boundary Violation:** Attempting to add 11 items (System displays maximum order limit warning).

3. Decision Table Methods:

Will be applied for the shipping fee calculation algorithm on D&R (Assuming free shipping for carts over 200 TL).

Conditions	Rule 1	Rule 2
Cart Total < 200 TL	True	False
Cart Total $\geq$ 200 TL	False	True
Actions		
Add Shipping Fee	Yes	No
Apply Free Shipping	No	Yes

4. Use-Case Analysis:

Guest User Scenario: Enter the site → Search for “Lord of the Rings” → Click on the first book → Click Add to Cart button → Navigate to Cart tab to verify the item is added.

3.6 Entry and exit criteria

Entry Criteria:

- Complete installation of all test environments and software (IntelliJ, Maven, Selenium WebDriver, JUnit6) on team members' computers.
- The target website (D&R) being online and accessible.
- Test cases being approved and documented.

Exit Criteria:

- Successful execution of all written JUnit tests.
- Successful conversion and error-free execution of at least 2 Selenium GUI tests into Java code.
- Completion of Equivalence and Boundary tests for all in-scope functionalities.

3.7 Test completion criteria

The test execution process will be considered complete when the Code Coverage ratio reaches a minimum of 80% on JUnit, and the “Test Completion Report” is finalized before the May 4th presentation.

3.8 Metrics to be collected

The following quality metrics will be collected throughout the process:

- Total Number of Test Cases / Number of Executed Tests.
- Passed and Failed Test Ratios.
- Line and Method Coverage Percentage (via IntelliJ).
- Mutation Score (via PIT plugin).

3.9 Test data requirements

Sanitized and anonymous test data will be used for D&R tests. In accordance with ISO/IEC 25012 standards, our test data includes:

- **Search Test Data:** Valid book names, invalid searches containing special characters.
- **Price Test Data:** Random numeric values such as 0, 50, 100, -5, 99999.

Real credit card or member information will not be used due to personal data privacy and security barriers.

3.10 Test environment requirements

The required hardware and software configuration for the tests is as follows:

- **Hardware:** Personal computers with a minimum of 8GB RAM.
- **IDE (Development Environment):** IntelliJ Community Edition.
- **Compiler/Runner:** Oracle JDK 17+.
- **Browser Automation:** Google Chrome, Selenium IDE extension, Maven Selenium-Java Library (v4.40.0).
- **Libraries:** JUnit6 (org.junit.jupiter:junit-jupiter:6.0.2).
- **Static Analysis:** PMDPlugin, CheckStyle-IDE.

3.11 Testing activities and estimates

The planned testing activities are as follows:

Activity	Estimated Duration	Responsible Person
Environment Setup (IntelliJ, Maven)	2 Days	Entire Team
Creation of the Test Plan	3 Days	Nazif Tosun
Recording Scenarios with Selenium IDE	4 Days	Mert Alp Barakazi
Converting GUI Tests to Java Code	5 Days	Mert Alp Barakazi
Writing JUnit Tests for Library-DB	6 Days	Levent Özada
Mutation Analysis & Coverage Reporting	3 Days	Muhammet Duran
Preparation of Final Test Report	4 Days	Entire Team

4 Staffing

4.1 General

This project will be executed by a test team of members with an equal distribution of workload, in accordance with software quality assurance principles.

4.2 Roles and responsibilities

Team members and their job descriptions are as follows:

- **Mert Alp Barakazi (Test Automation Engineer):** Responsible for recording scenarios on D&R using Selenium IDE and converting at least 2 of these recordings into executable Java WebDriver code.
- **Levent Özada (Unit Test Specialist):** Responsible for designing and writing unit tests for the Library-DB-Express project using the JUnit6 library.
- **Nazif Tosun (Test Analyst & Documentation):** Responsible for creating the Test Plan template, documenting the test design techniques (Equivalence, Boundary, etc.), and compiling the final Test Completion Report.

- **Muhammet Duran (Quality and Mutation Expert):** Responsible for extracting the Coverage ratios of the written codes, executing mutation tests via the PIT plugin, and detecting potential static code analysis errors.

4.3 Schedule

General testing schedule:

- **April, 3rd Week:** Environment setup, finalizing D&R test scenarios, and uploading the Test Plan to the system.
- **April, 4th Week:** Writing D&R Selenium automation codes and completing Library-DB-Express JUnit tests.
- **May, 1st Week:** Executing mutation tests, collecting metrics, preparing the Test Completion Report, and rehearsing for the laboratory presentation on May 4th.